Vertical CNC Machining Centre (CNC-MILL) Equipment Manual and SOP

STANDARD OPERATING PROCEDURE AND MAINTENANCE MANUAL Asset Class: CNC-MILL (Vertical CNC Machining Centre) Document ID: SOP-VPW-CNC-042 Revision: 04 Effective Date: 15 October 2024 Plant Location: Vindhya Precision Works, Plant Unit 3, Machining Bay B

1. PURPOSE AND APPLICABILITY

1.1 This document defines the mandatory operating envelopes, preventive maintenance schedules, fault diagnosis protocols, torque values, and spare parts management for asset class CNC-MILL within Vindhya Precision Works.

1.2 Compliance with this document is mandatory for all Maintenance Engineering, Toolroom, and Production Operations personnel.

1.3 Responsible Trade: Mechanical is primary, with Electrical assigned in direct support for all electrical interlock, drive, and control cabinet interventions.

2. HAZARDOUS ENERGY ISOLATION

2.1 Prior to performing any corrective maintenance, tool change of non-standard configuration, or preventive maintenance requiring physical enclosure entry, technicians shall execute complete Lockout/Tagout (LOTO) procedures in accordance with Vindhya Precision Works safety standard EHS-LOTO-01.

2.2 Hazardous energy sources identified for asset class CNC-MILL:

* Electrical: 415V AC, 3-phase supply via main rotary isolator switch QF1, located on the rear of the machine control cabinet. Control voltage (24V DC) must be verified dead at terminal block X10.
* Pneumatic: 6.0 bar compressed air supply via main air service unit shut-off valve V-AIR-01, equipped with an exhaust port padlock loop.

3. NORMAL OPERATING RANGES AND CONDITION MONITORING THRESHOLDS

3.1 Healthy units within asset class CNC-MILL shall operate strictly within the baseline parameters defined in Table 3.1.

Table 3.1: Operating Envelopes and Alarm Setpoints for Asset Class CNC-MILL

Parameter	Baseline Healthy Range	Warning Setpoint	Trip / Interlock Setpoint	Monitored Status
Spindle Vibration (RMS)	1.2 to 4.0 mm/s	4.5 mm/s RMS	7.1 mm/s RMS	Continuous (CMS Channel 1)
Spindle/Housing Surface Temp	38 °C to 66 °C	72 °C	86 °C	Continuous (CMS Channel 2)
Hydraulic/Pneumatic Pressure	Not Applicable	Not Applicable	Not Applicable	No monitored process pressure on this asset class
Spindle Motor Current (Normal Load)	14.0 A to 38.5 A	45.0 A (10s delay)	52.0 A (Immediate Trip)	Polled via PLC Register R-402

3.2 Vibration Warning (4.5 mm/s RMS): Generates an amber advisory on the shop floor SCADA terminal. The machine may complete the active part program, but subsequent cycles are inhibited until a mechanical inspection is logged.
3.3 Vibration Trip (7.1 mm/s RMS): Initiates an immediate hardware emergency stop, cutting servo and spindle power within 120 milliseconds.
3.4 Surface Temperature Warning (72 °C): Triggers a coolant flow verification flag on the operator HMI.
3.5 Surface Temperature Trip (86 °C): Initiates a controlled feed-hold, spindle deceleration, and cycle abort.

4. PREVENTIVE MAINTENANCE SCHEDULE

4.1 The base preventive maintenance interval for asset class CNC-MILL is 500 running hours (measured via PLC hour-meter register HR-01).
4.2 Technicians shall execute tasks in accordance with the running-hour milestone matrix detailed in Table 4.1.

Table 4.1: Preventive Maintenance Milestone Matrix

Interval	Task Description	Responsible Trade	Estimated Hours
Every 500 Hours	Replace Training filter (Part ID VPW-P-00076). Inspect Way lube reservoir levels. Check spindle taper runout. Clean cabinet ventilation grills.	Mechanical	2.5
Every 1,000 Hours	Replace Hour filter (Part ID VPW-P-00008). Inspect and grease axis ball screws (X, Y, Z axes) with Mobilith SHC 220. Verify axis backlash using laser interferometer.	Mechanical	4.0
Every 2,000 Hours	Full overhaul of pneumatic distribution blocks. Replace Drug valve (Part ID VPW-P-00021). Inspect Report coupling (Part ID VPW-P-00067) for rotational play.	Mechanical & Electrical	8.0
Every 4,000 Hours	Replace Investment bearing (Part ID VPW-P-00054) assemblies on primary axes. Calibrate Still sensor (Part ID VPW-P-00051) and Dark sensor (Part ID VPW-P-00079) against NIST-traceable standards. Overhaul Remain seal (Part ID VPW-P-00006) on spindle head.	Mechanical & Electrical	16.0

5. FAULT-CODE DIAGNOSTICS AND CORRECTIVE ACTIONS

5.1 When the CNC control unit or CMS registers a fault, technicians shall reference Table 5.1 for diagnosis and mandatory remediation.

Table 5.1: Fault-Code Prose Description and Corrective Action Matrix

Fault Condition / Code	Description	Probable Cause	Corrective Action
ERR-CNC-101	Spindle Over-Temperature Trip	Coolant pump cavitation or heat exchanger fouling.	Verify fluid flow; clean heat exchanger fins; inspect Still sensor (Part ID VPW-P-00051) wiring integrity.
ERR-CNC-102	Excessive Vibration Interlock	Spindle bearing micro-spalling or tool unbalance.	Stop machine; inspect toolholder balance; check Investment bearing (Part ID VPW-P-00054) condition via CMS history.
ERR-CNC-103	Lube Pressure Fault	Lubrication distributor blockage or pump failure.	Inspect Way lube lines; replace Training filter (Part ID VPW-P-00076); verify pump cycle timer in PLC.
ERR-CNC-104	Axis Servo Drive Overload	Mechanical binding on ball screw or way covers.	Isolate electrical power; manually check axis drag; inspect Report coupling (Part ID VPW-P-00067) alignment.
ERR-CNC-105	Pneumatic Clamping Failure	Insufficient clamping pressure or seal degradation.	Check incoming air supply (min 6.0 bar); replace Remain seal (Part ID VPW-P-00006) on drawbar cylinder.
ERR-CNC-106	Coolant Flow Interlock	Fluid starvation or heavy chip accumulation in sump.	Clean chip conveyor; replace Hour filter (Part ID VPW-P-00008); purge air pockets from fluid lines using Drug valve (Part ID VPW-P-00021).

6. TORQUE SPECIFICATIONS

6.1 All fasteners utilized during maintenance of asset class CNC-MILL shall be tightened using calibrated torque wrenches in accordance with Table 6.1.

Table 6.1: Fastener Torque Requirements

Fastener Application	Thread Size	Strength Class	Tightening Torque ($\pm 5\%$)
Spindle Cartridge Clamping Bolts	M12	12.9	85.0 Nm
Axis Ball Screw Bearing Locknuts	M20x1.5	10.9	150.0 Nm
Tool Changer Arm Mounting Bolts	M10	8.8	46.0 Nm
Way Cover Retaining Screws	M8	8.8	25.0 Nm
Motor Coupling Clamp Screws	M6	12.9	14.5 Nm

7. CONSUMABLE AND SPARE PARTS INVENTORY

7.1 Technicians shall exclusively use the following plant-inventory-controlled part numbers when servicing asset class CNC-MILL. No substitutions are permitted without written engineering authorization from the Maintenance Engineering Lead.

* VPW-P-00006 – Remain seal (Spindle drawbar assembly)
* VPW-P-00008 – Hour filter (Primary coolant filtration unit)
* VPW-P-00021 – Drug valve (Pneumatic purge and fluid line control valve)
* VPW-P-00051 – Still sensor (Thermal monitoring probe, spindle housing)
* VPW-P-00054 – Investment bearing (Primary axis support bearing set)
* VPW-P-00067 – Report coupling (Servo-to-ball screw torque transmission coupling)
* VPW-P-00076 – Training filter (Way lube metering unit inline filter)
* VPW-P-00079 – Dark sensor (Optical zero-position reference sensor)

8. SAFETY-CRITICAL CORRECTIVE ACTIONS

8.1 The following corrective actions are designated as Safety-Critical. They require mandatory execution of Lockout/Tagout (LOTO) per Section 2 and the issuance of a confined-space or hot-work permit (as applicable) before work commences:

1. Replacement or mechanical adjustment of Investment bearing (Part ID VPW-P-00054) assemblies.
2. Servicing, alignment, or replacement of Report coupling (Part ID VPW-P-00067) under dynamic load conditions.
3. Overhaul or resealing of the spindle cartridge involving Remain seal (Part ID VPW-P-00006).
4. Any direct intervention within the main 415V electrical control enclosure involving the servo drives or main breaker QF1.